

① Radialno polarizirana krogla



$$\vec{P} = k\vec{r}$$

a) $\rho_v, \sigma_v, e_{TOT} = ?$
 b) $\vec{E}(\vec{r}) = ?$

$$\rho_v = -\nabla \cdot \vec{P} = -3k$$

$$\sigma_v = \vec{P} \cdot \hat{n} = k\vec{r} \cdot \frac{\vec{r}}{r} = kr|_{r=a} = ka$$

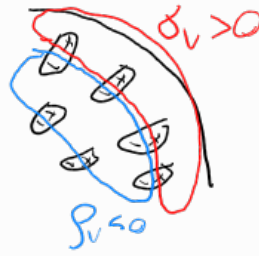
$$e_{TOT} = \int \rho_v dV + \int \sigma_v dS = \rho_v \int dV + \sigma_v \int dS = -3k \frac{4\pi a^3}{3} + ka 4\pi a^2 = 0$$

$$\rho = \rho_v + \rho_p = \nabla \cdot \epsilon_0 \vec{E}$$

$$\int \rho_v dV = \oint \epsilon_0 \vec{E} \cdot d\vec{S} = \epsilon_0 \vec{E}(r) S(r) = \epsilon_0 \vec{E}(r) 4\pi r^2$$

$$-3k \frac{4\pi r^3}{3}$$

$$\vec{E}(r) = -\frac{k}{\epsilon_0} r \rightarrow \vec{E}(\vec{r}) = -\frac{\vec{P}}{\epsilon_0}$$



② Homogena polarizirana krogla

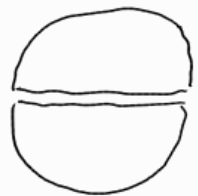


a) $U(r, \vartheta) = ?$ Povsod

$$\vec{E}_{not} = ?$$

$$\vec{P} = \begin{bmatrix} 0 \\ 0 \\ P \end{bmatrix} \rightarrow \rho_v = -\frac{\partial P}{\partial z} = 0$$

$$\hat{n} = \begin{bmatrix} \sin \vartheta \cos \varphi \\ \sin \vartheta \sin \varphi \\ \cos \vartheta \end{bmatrix}$$



b) $\vec{E} = ?$ (v špranji)

$$\sigma_v = P \cos \vartheta$$

$$U(r, \vartheta) = \sum_{l=0}^{\infty} (A_l r^l + B_l r^{-(l+1)}) P_l(\cos \vartheta)$$

$$U(r, \vartheta) = (A_1 r + B_1 r^{-2}) \cos \vartheta = \begin{cases} A_1 r \cos \vartheta; & r < a \\ B_1 r^{-2} \cos \vartheta; & r > a \end{cases}$$

RP1: zveznost U

RP2: $\frac{\sigma_v}{\epsilon_0} = E_{zvn}^{\perp} - E_{not}^{\perp}$

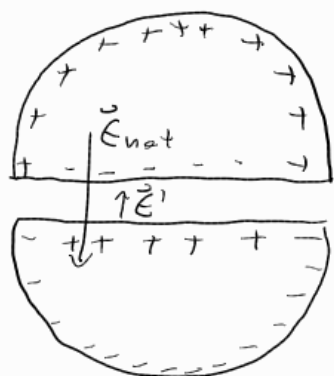
RP1: $A_1 a \cos \vartheta = B_1 a^{-2} \cos \vartheta \quad \left[A_1 = \frac{B_1}{a^3} \right]$

RP2: $\frac{P \cos \vartheta}{\epsilon_0} = -[B_1 (-2) a^{-3} - A_1] \cos \vartheta = B_1 a^{-3} (2+1) = 3B_1 a^{-3} \rightarrow B_1 = \frac{Pa^3}{3\epsilon_0}$

$$\rightarrow U(r, \vartheta) = \begin{cases} \frac{P}{3\epsilon_0} r \cos \vartheta ; r < a \\ \frac{P}{3\epsilon_0} \frac{a^3}{r^2} \cos \vartheta ; r > a \end{cases}$$

$$E_{\text{not}} = -\nabla U = -\nabla \frac{P}{3\epsilon_0} r \cos \vartheta = -\nabla \frac{P}{3\epsilon_0} z = \begin{bmatrix} 0 \\ 0 \\ -\frac{P}{3\epsilon_0} \end{bmatrix} = -\frac{\vec{P}}{3\epsilon_0} \quad \frac{1}{3} \text{ depolarizacijski faktor}$$

b)

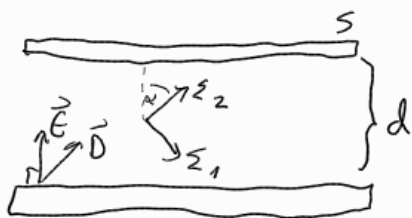


$$\sigma'_v = P \text{ (na površini špranje)}$$

$$E' = \sigma'_v / \epsilon_0$$

$$\vec{E} = \vec{E}_{\text{not}} + \vec{E}' = \frac{P}{\epsilon_0} - \frac{P}{3\epsilon_0} = -\frac{2}{3} \frac{P}{\epsilon_0}$$

③. ploščati kond. z anizotropnim dielektrikom



$$C = \frac{Q}{U} = ?$$

$$\rho_p = \nabla \cdot (\underbrace{\epsilon_0 \vec{E} + \vec{P}}_{\vec{D}}) \quad \text{Dielektrik: } \vec{D} = \underline{\underline{\epsilon}} \epsilon_0 \vec{E}$$

$$\vec{E} = \begin{bmatrix} Q \\ \epsilon \end{bmatrix} = \begin{bmatrix} 0 \\ U/d \end{bmatrix}$$

$$\text{RP1: } \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = 0$$

$$\text{RP2: } \nabla \cdot \vec{D} = \rho_p \quad / \int dV \text{ po celi plošči}$$

$$\int \rho_p dV = Q = \int \nabla \cdot \vec{D} dV = \int \vec{D} d\vec{S} = (D_{\text{not}}^{\perp} - \cancel{D_{\text{zuvn}}^{\perp}}) S = D^{\perp} S$$

$$\underline{\underline{\epsilon}} = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \epsilon_1 & 0 \\ 0 & \epsilon_2 \end{bmatrix} \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} = \dots = \begin{bmatrix} \epsilon_1 \cos^2 \alpha + \epsilon_2 \sin^2 \alpha & (\epsilon_2 - \epsilon_1) \sin \alpha \cos \alpha \\ (\epsilon_2 - \epsilon_1) \sin \alpha \cos \alpha & \epsilon_1 \sin^2 \alpha + \epsilon_2 \cos^2 \alpha \end{bmatrix}$$

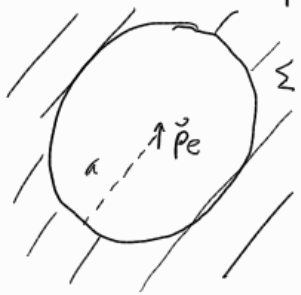
$$\vec{D} = \begin{bmatrix} D^{\parallel} \\ D^{\perp} \end{bmatrix} = \begin{bmatrix} D^{\parallel} \\ \epsilon_1^{\perp} S \end{bmatrix} = \epsilon_0 \underline{\underline{\epsilon}} \vec{E} = \epsilon_0 \begin{bmatrix} \phantom{D^{\parallel}} \\ U/d \end{bmatrix}$$

$$\frac{Q}{S} = \epsilon_0 \frac{U}{d} [\epsilon_1 \sin^2 \alpha + \epsilon_2 \cos^2 \alpha]$$

↙ kond. brez dielektrika

$$C = \frac{Q}{U} = \epsilon_0 \frac{S}{d} [\epsilon_1 \sin^2 \alpha + \epsilon_2 \cos^2 \alpha] = \underline{\underline{C_0 [\epsilon_1 \sin^2 \alpha + \epsilon_2 \cos^2 \alpha]}}$$

4. Točkast dipol v krogelni votlini dielektrika



a) $U(r, \vartheta) = ?$ Povsod

$p'_e = ?$ kot je videti iz dielektrika

b) $\sigma_v(\vartheta) = ?$

$$U_{DIP} = \frac{\vec{p}_e \cdot \vec{r}}{4\pi\epsilon_0 r^3} = \frac{p_e}{4\pi\epsilon_0} \frac{\cos\vartheta}{r^2}$$

$$\rightarrow U = (A_1 r + B_1 r^{-2}) \cos\vartheta = \begin{cases} \frac{p_e}{4\pi\epsilon_0} \frac{\cos\vartheta}{r^2} + A_1 r \cos\vartheta; & r < a \\ B_1 \frac{\cos\vartheta}{r^2}; & r > a \end{cases}$$